

ASE230 LMS API - How to Put It Online

How to Make Your Website Live

Made by: Bishal Bhandari

Class: ASE230 - Server-Side Programming

Date: October 2025

Getting Ready to Go Live



What you need before putting your website online

What you need:

- ☒ PHP 7.4+ with MySQL support
- ☒ MySQL 5.7+ or MariaDB 10.3+
- ☒ Web server (Apache/Nginx)
- ☒ SSL certificate for HTTPS
- ☒ Domain name configured

What your server needs:

- **PHP Extensions:** PDO, JSON, OpenSSL
- **Memory:** At least 256MB PHP memory limit

Setting Up Your Web Server

How to configure your web server

Apache Configuration

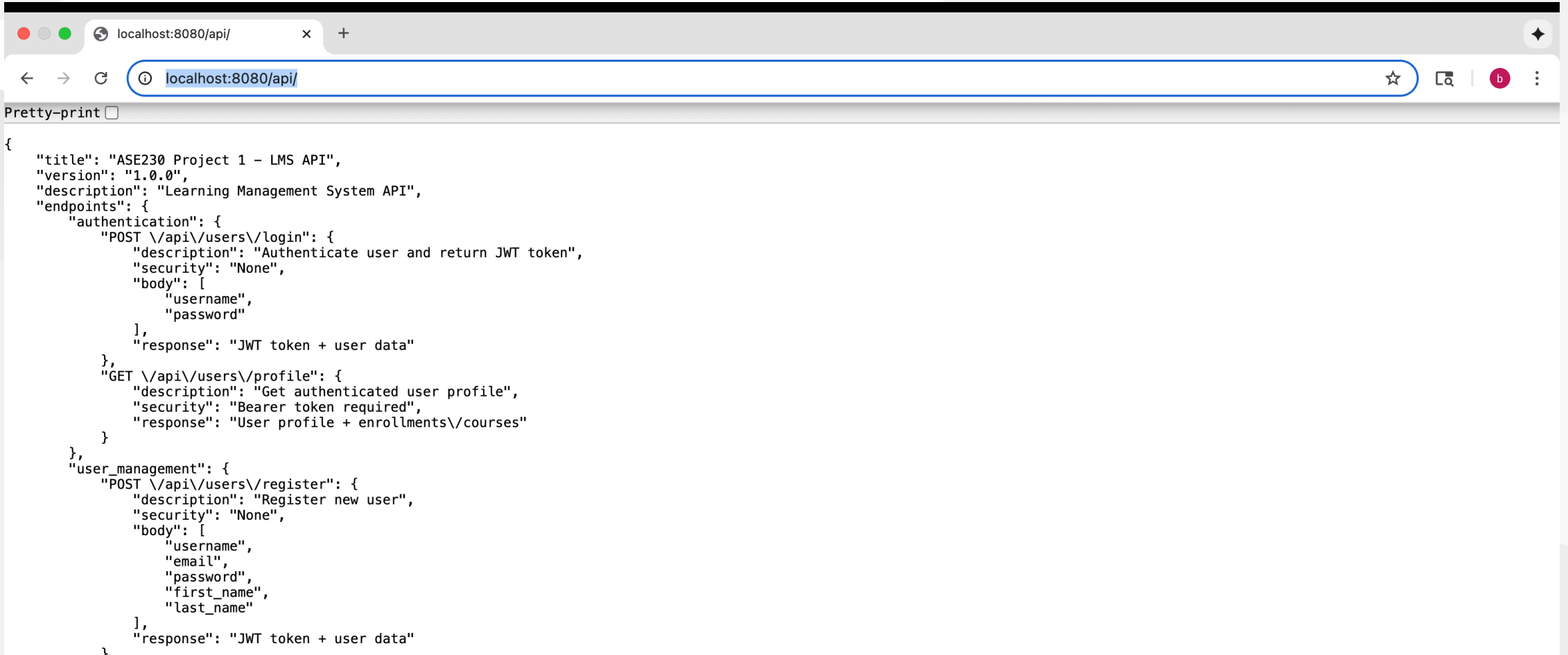
```
<VirtualHost *:80>
    ServerName your-domain.com
    DocumentRoot /var/www/html/ase230-lms

    <Directory /var/www/html/ase230-lms>
        AllowOverride All
        Require all granted
    </Directory>

    # Send everyone to HTTPS
    RewriteEngine On
    RewriteCond %{HTTPS} off
    RewriteRule ^(.*)$ https://%{HTTP_HOST}%{REQUEST_URI} [L,R=301]
</VirtualHost>
```

NGINX Working Successfully

Proof that NGINX is serving your API



A screenshot of a web browser window with the address bar showing `localhost:8080/api/`. The page content displays a JSON response from the API, which is formatted using a "Pretty-print" button. The JSON data describes the "ASE230 Project 1 - LMS API" with version "1.0.0" and lists several endpoints including authentication, user profile retrieval, and user registration.

```
{
  "title": "ASE230 Project 1 - LMS API",
  "version": "1.0.0",
  "description": "Learning Management System API",
  "endpoints": {
    "authentication": {
      "POST \\/api\\/users\\/login": {
        "description": "Authenticate user and return JWT token",
        "security": "None",
        "body": [
          "username",
          "password"
        ],
        "response": "JWT token + user data"
      },
      "GET \\/api\\/users\\/profile": {
        "description": "Get authenticated user profile",
        "security": "Bearer token required",
        "response": "User profile + enrollments\\/courses"
      }
    },
    "user_management": {
      "POST \\/api\\/users\\/register": {
        "description": "Register new user",
        "security": "None",
        "body": [
          "username",
          "email",
          "password",
          "first_name",
          "last_name"
        ],
        "response": "JWT token + user data"
      }
    }
  }
}
```

Setting Up Your Database

How to set up MySQL database

1. Create Database

```
CREATE DATABASE ase230_lms CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;  
CREATE USER 'lms_user'@'localhost' IDENTIFIED BY 'secure_password';  
GRANT ALL PRIVILEGES ON ase230_lms.* TO 'lms_user'@'localhost';  
FLUSH PRIVILEGES;
```

2. Import Your Data

```
mysql -u lms_user -p ase230_lms < database_schema.sql
```

3. Update Your Settings

```
// config/db.php
```

Keeping Things Safe



How to make your website secure

1. Environment Variables

```
# .env file
DB_HOST=localhost
DB_NAME=ase230_lms
DB_USER=lms_user
DB_PASS=secure_password
JWT_SECRET=your_super_secret_jwt_key_here
API_BASE_URL=https://your-domain.com/api
```

2. File Permissions

```
# Set proper permissions
chmod 755 /var/www/html/ase230-lms
chmod 644 /var/www/html/ase230-lms/config/db.php
```

Setting Up HTTPS

How to get free SSL certificate

Let's Encrypt (Free SSL)

```
# Install Certbot
sudo apt-get install certbot python3-certbot-apache

# Get certificate
sudo certbot --apache -d your-domain.com

# Auto-renewal
sudo crontab -e
# Add: 0 12 * * * /usr/bin/certbot renew --quiet
```

SSL Best Practices

- TLS Version: At least TLS 1.2

Making It Faster

How to make your website run better

1. PHP Configuration

```
; php.ini optimizations
memory_limit = 512M
max_execution_time = 30
upload_max_filesize = 10M
post_max_size = 10M
opcache.enable = 1
opcache.memory_consumption = 128
```

2. Database Optimization

```
-- Add indexes for better performance
CREATE INDEX idx_users_email ON users(email);
CREATE INDEX idx_courses_semester_year ON courses(semester, year);
```


Watching Your Website



How to keep track of what's happening

1. Error Logging

```
// Log configuration
ini_set('log_errors', 1);
ini_set('error_log', '/var/log/php/ase230-lms-errors.log');
```

2. Application Logs

```
// Custom logging function
function logAPI($message, $level = 'INFO') {
    $timestamp = date('Y-m-d H:i:s');
    $logEntry = "[ $timestamp ] [ $level ] $message" . PHP_EOL;
    file_put_contents('/var/log/ase230-lms/app.log', $logEntry, FILE_APPEND);
}
```

Backing Up Your Data

How to save your data safely

1. Database Backups

```
#!/bin/bash
# Daily backup script
DATE=$(date +%Y%m%d_%H%M%S)
mysqldump -u lms_user -p ase230_lms > /backups/lms_backup_$(date +%Y%m%d_%H%M%S).sql
gzip /backups/lms_backup_$(date +%Y%m%d_%H%M%S).sql

# Keep only last 30 days
find /backups -name "lms_backup_*.sql.gz" -mtime +30 -delete
```

2. Application Backups

```
# Backup application files
tar -czf /backups/app backup $(date +%Y%m%d).tar.gz /var/www/html/ase230-lms
```

Deployment Process

Automated Deployment

1. Git-based Deployment

```
#!/bin/bash
# deploy.sh script
cd /var/www/html/ase230-lms
git pull origin main
composer install --no-dev --optimize-autoloader
php artisan config:cache
php artisan route:cache
```

2. CI/CD Pipeline

```
# .github/workflows/deploy.yml
name: Deploy to Production
on:
```

Testing in Production



Production Testing

1. Smoke Tests

```
# Basic functionality tests
curl -f https://your-domain.com/api/courses
curl -f https://your-domain.com/api/users/register -X POST
```

2. Load Testing

```
# Using Apache Bench
ab -n 1000 -c 10 https://your-domain.com/api/courses
```

3. Security Testing

- **SSL Labs Test:** Check SSL configuration

Maintenance & Updates

Ongoing Maintenance

1. Regular Updates

- **PHP Updates:** Keep PHP version current
- **Security Patches:** Apply OS and application patches
- **Dependencies:** Update third-party libraries

2. Performance Monitoring

- **Response Times:** Monitor API response times
- **Error Rates:** Track application error rates
- **Resource Usage:** Monitor server resources

Troubleshooting Guide

Common Issues & Solutions

1. Database Connection Issues

```
# Check database connectivity
mysql -u lms_user -p -h localhost ase230_lms -e "SELECT 1"

# Check PHP PDO extension
php -m | grep pdo
```

2. SSL Certificate Problems

```
# Check certificate validity
openssl x509 -in /path/to/certificate.crt -text -noout

# Test SSL connection
openssl s_client -connect your-domain.com:443
```

Deployment Complete!

Production Ready Features

- Secure HTTPS Configuration
- Database Optimization
- Performance Monitoring
- Automated Backups
- Error Handling
- Security Hardening

Support & Maintenance

- **24/7 Monitoring:** Automated health checks
- **Regular Backups:** Daily automated backups
- **Security Updates:** Regular security patches

Github

Github Repo

Repository: <https://github.com/BsalBhandari/ase230-bhandari-project1>

Appendix: Deployment Commands

Quick Reference Commands

Database Operations

```
# Backup database
mysqldump -u lms_user -p ase230_lms > backup.sql

# Restore database
mysql -u lms_user -p ase230_lms < backup.sql

# Check database status
mysqladmin -u lms_user -p status
```

Application Management

```
# Check PHP configuration
php -i | grep -E "(memory_limit|max_execution_time)"
```